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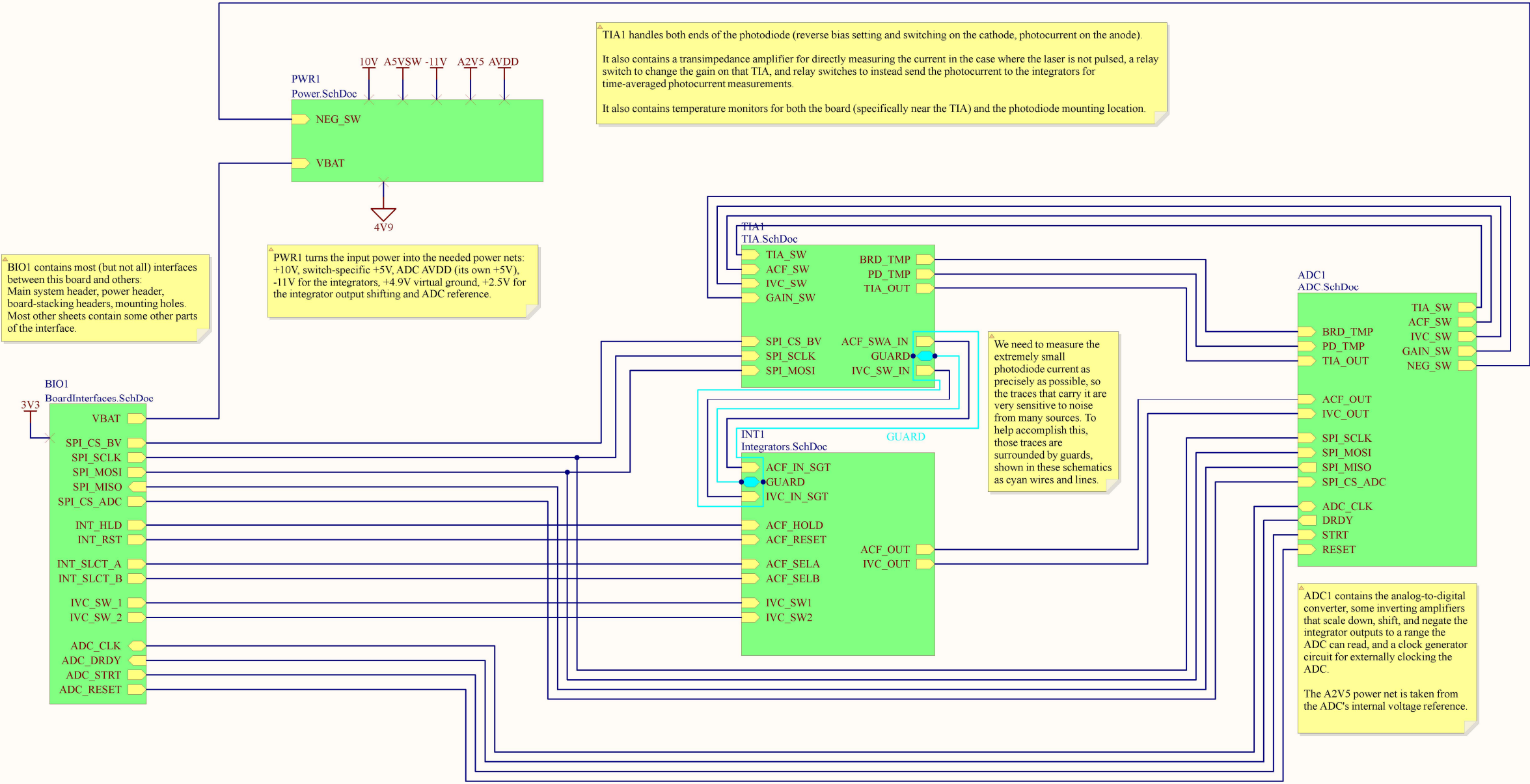
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Changes from V26B include:

- Added TIA isolation relays and associated control lines to prevent amplifier contention
- Add 2:4 Decoder to allow for one extra IO relay control and enforce amplifier mutual exclusivity
- Rework power-on sequencing to ensure that ADC AVDD is on before input voltages on its lines rise above allowable thresholds
- Add input anti-aliasing filters for all critical inputs (also provides over/undervoltage protection)
- Fix thermistors lacking any bias
- Updated solder to bring it in-line with V26B baseline
- Various little fixes
 - Fix fast-start for 10V
 - 5V adjustment resistors removed
 - Drop TIA max voltage from 10V to 5V to prevent overvoltage events in case of negative photocurrent

THIS REVISION, V26_, WAS SENT FOR FABRICATION ON ____.

Compatible board: Soldier: ____



TIA1 handles both ends of the photodiode (reverse bias setting and switching on the cathode, photocurrent on the anode).

It also contains a transimpedance amplifier for directly measuring the current in the case where the laser is not pulsed, a relay switch to change the gain on that TIA, and relay switches to instead send the photocurrent to the integrators for time-averaged photocurrent measurements.

It also contains temperature monitors for both the board (specifically near the TIA) and the photodiode mounting location.

BIO1 contains most (but not all) interfaces between this board and others: Main system header, power header, board-stacking headers, mounting holes. Most other sheets contain some other parts of the interface.

PWR1 turns the input power into the needed power nets: +10V, switch-specific +5V, ADC AVDD (its own +5V), -11V for the integrators, +4.9V virtual ground, +2.5V for the integrator output shifting and ADC reference.

We need to measure the extremely small photodiode current as precisely as possible, so the traces that carry it are very sensitive to noise from many sources. To help accomplish this, those traces are surrounded by guards, shown in these schematics as cyan wires and lines.

ADC1 contains the analog-to-digital converter, some inverting amplifiers that scale down, shift, and negate the integrator outputs to a range the ADC can read, and a clock generator circuit for externally clocking the ADC.

The A2V5 power net is taken from the ADC's internal voltage reference.

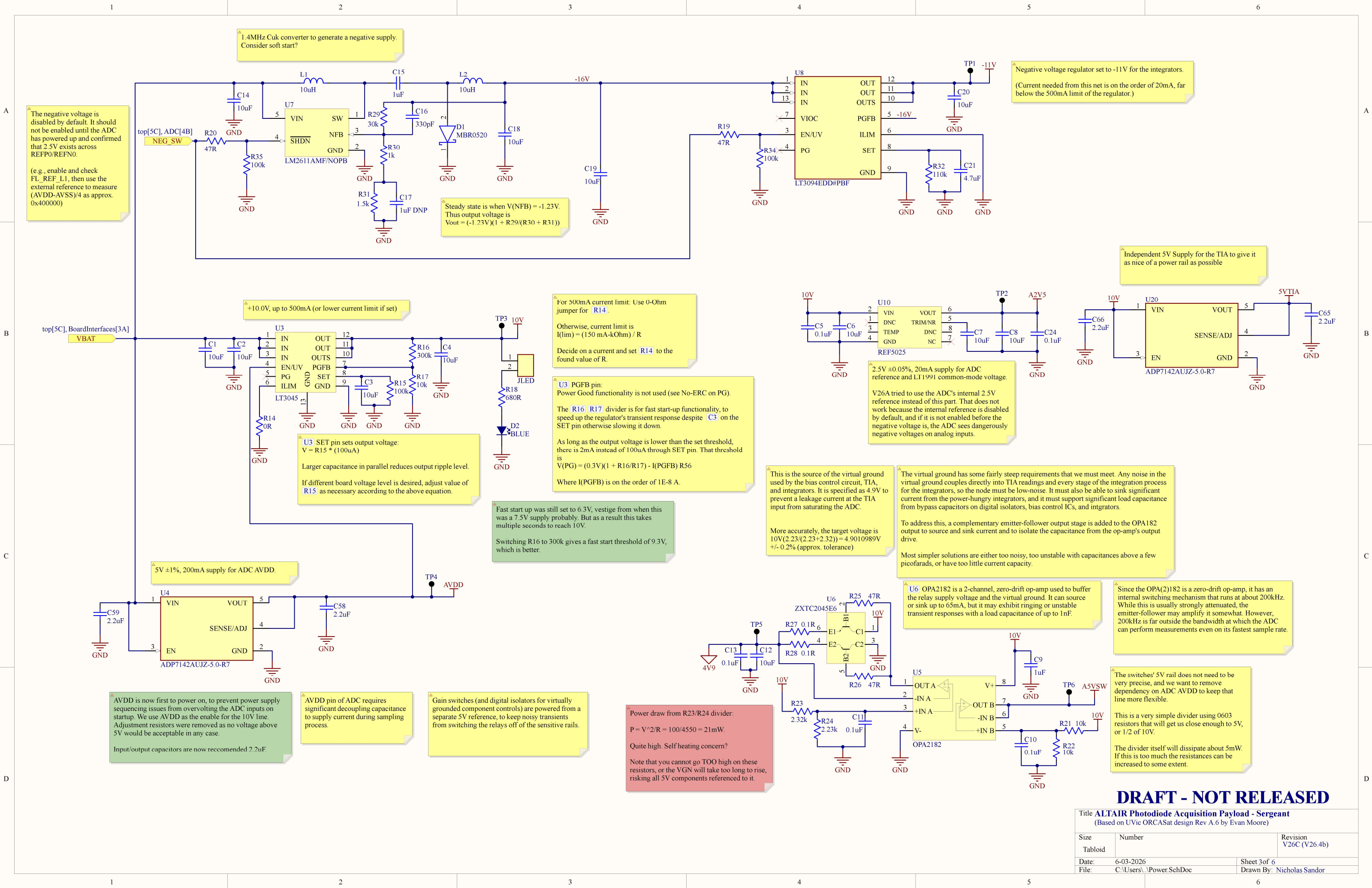
INT1 contains two integrator circuits.

One works with the ACF2101 dual-channel switched integrator, which is the primary measurement device for photocurrents from both the Sergeant board and the stacked Soldier board, connected through spring-loaded pogo pins and pads.

The other works with the IVC102 single-channel switched integrator, which serves to validate the values given by the ACF2101. The Soldier board has its own IVC102 as well.

DRAFT - NOT RELEASED

Title ALTAIR Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)		
Size Tabloid	Number	Revision V26C (V26.4b)
Date:	6-03-2026	Sheet 1 of 6
File:	C:\Users\...\top.SchDoc	Drawn By: Nicholas Sandor



The negative voltage is disabled by default. It should not be enabled until the ADC has powered up and confirmed that 2.5V exists across REFPO/REFNO.

(e.g., enable and check FL_REF_L1, then use the external reference to measure (AVDD-AVSS)/4 as approx. 0x400000)

1.4MHz Cuk converter to generate a negative supply. Consider soft start?

Steady state is when $V(NFB) = -1.23V$. Thus output voltage is $V_{out} = (-1.23V)(1 + R_{29}/(R_{30} + R_{31}))$

Negative voltage regulator set to -11V for the integrators.

(Current needed from this net is on the order of 20mA, far below the 500mA limit of the regulator.)

Independent 5V Supply for the TIA to give it as nice of a power rail as possible

For 500mA current limit: Use 0-Ohm jumper for R14.

Otherwise, current limit is $I(lim) = (150\text{ mA-kOhm}) / R$

Decide on a current and set R14 to the found value of R.

U3 PGFB pin: Power Good functionality is not used (see No-ERC on PG).

The R16 R17 divider is for fast start-up functionality, to speed up the regulator's transient response despite C3 on the SET pin otherwise slowing it down.

As long as the output voltage is lower than the set threshold, there is 2mA instead of 100uA through SET pin. That threshold is $V(PG) = (0.3V)(1 + R_{16}/R_{17}) - I(PGFB) R_{56}$

Where $I(PGFB)$ is on the order of 1E-8 A.

Fast start up was still set to 6.3V, vestige from when this was a 7.5V supply probably. But as a result this takes multiple seconds to reach 10V.

Switching R16 to 300k gives a fast start threshold of 9.3V, which is better.

This is the source of the virtual ground used by the bias control circuit, TIA, and integrators. It is specified as 4.9V to prevent a leakage current at the TIA input from saturating the ADC.

More accurately, the target voltage is $10V(2.23/(2.23+2.32)) = 4.9010989V \pm 0.2\%$ (approx. tolerance)

The virtual ground has some fairly steep requirements that we must meet. Any noise in the virtual ground couples directly into TIA readings and every stage of the integration process for the integrators, so the node must be low-noise. It must also be able to sink significant current from the power-hungry integrators, and it must support significant load capacitance from bypass capacitors on digital isolators, bias control ICs, and integrators.

To address this, a complementary emitter-follower output stage is added to the OPA182 output to source and sink current and to isolate the capacitance from the op-amp's output drive.

Most simpler solutions are either too noisy, too unstable with capacitances above a few picofarads, or have too little current capacity.

Since the OPA(2)182 is a zero-drift op-amp, it has an internal switching mechanism that runs at about 200kHz. While this is usually strongly attenuated, the emitter-follower may amplify it somewhat. However, 200kHz is far outside the bandwidth at which the ADC can perform measurements even on its fastest sample rate.

The switches' 5V rail does not need to be very precise, and we want to remove dependency on ADC AVDD to keep that line more flexible.

This is a very simple divider using 0603 resistors that will get us close enough to 5V, or 1/2 of 10V.

The divider itself will dissipate about 5mW. If this is too much the resistances can be increased to some extent.

Power draw from R23/R24 divider: $P = V^2/R = 100/4550 = 21mW$.

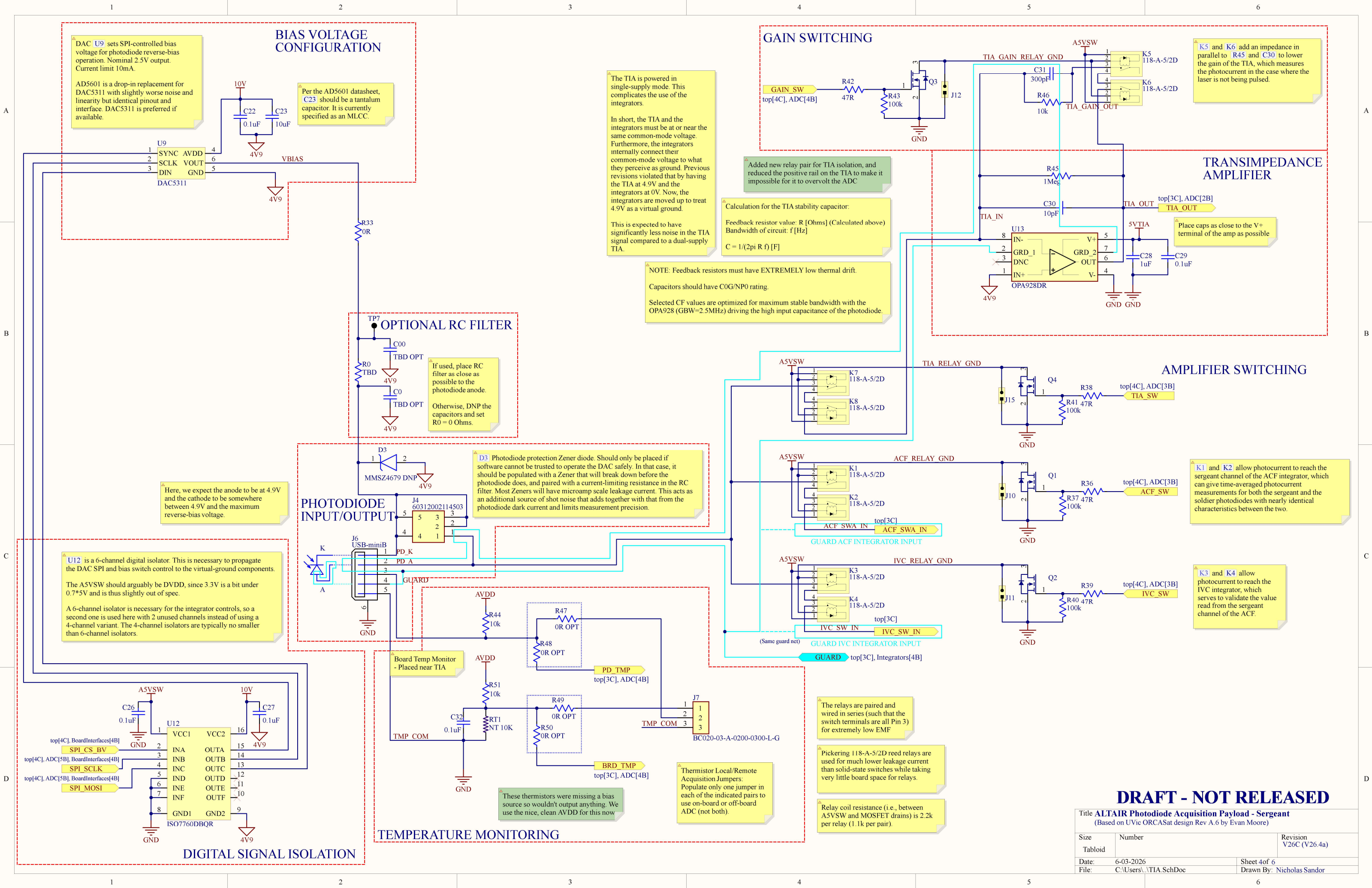
Quite high. Self heating concern?

Note that you cannot go TOO high on these resistors, or the VGN will take too long to rise, risking all 5V components referenced to it.

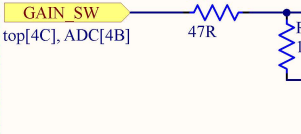
DRAFT - NOT RELEASED

Title ALTAIR Photodiode Acquisition Payload - Sergeant
(Based on Uvic ORCASat design Rev A.6 by Evan Moore)

Size	Number	Revision
Tabloid		V26C (V26.4b)
Date:	6-03-2026	Sheet 3 of 6
File:	C:\Users\...\Power.SchDoc	Drawn By: Nicholas Sandor



GAIN SWITCHING

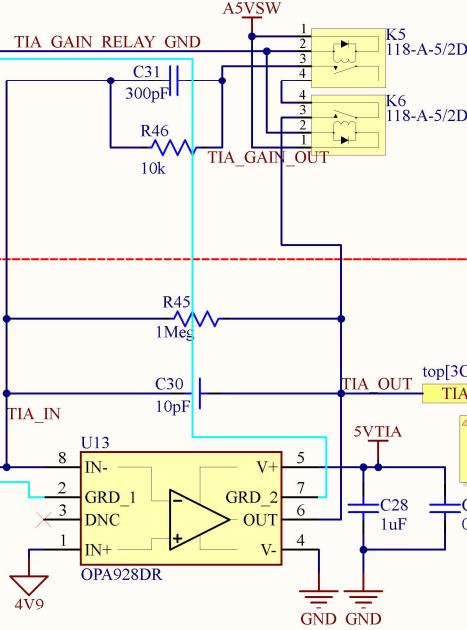


Added new relay pair for TIA isolation, and reduced the positive rail on the TIA to make it impossible for it to overvolt the ADC

Calculation for the TIA stability capacitor:
Feedback resistor value: R [Ohms] (Calculated above)
Bandwidth of circuit: f [Hz]
 $C = 1/(2\pi R f)$ [F]

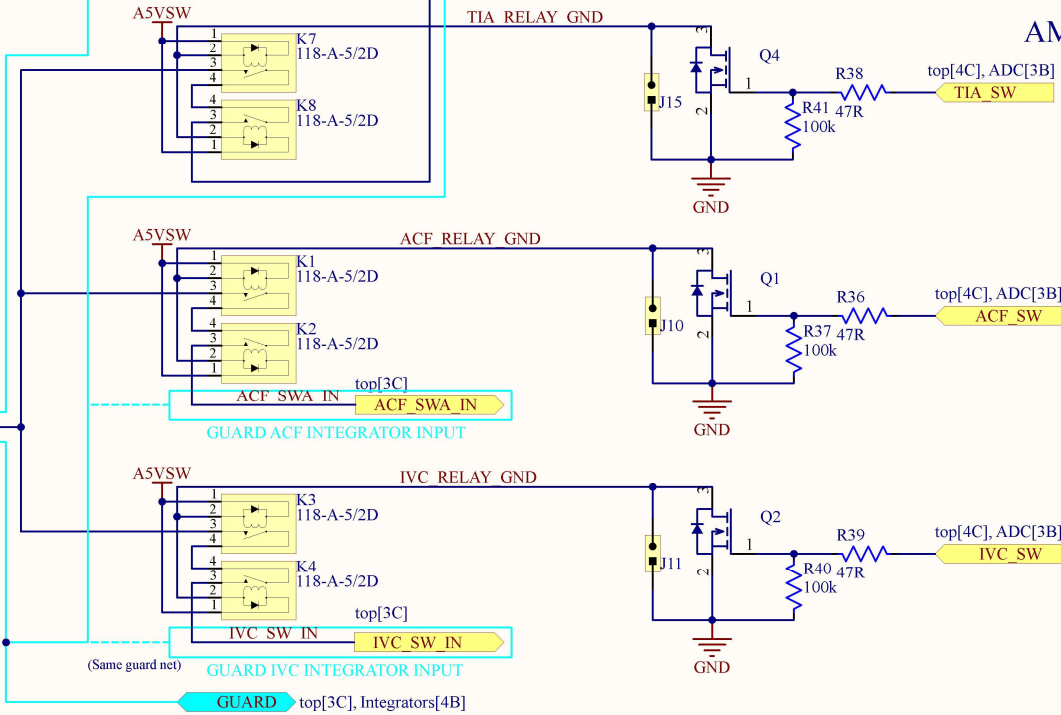
NOTE: Feedback resistors must have EXTREMELY low thermal drift.
Capacitors should have COG/NP0 rating.
Selected CF values are optimized for maximum stable bandwidth with the OPA928 (GBW=2.5MHz) driving the high input capacitance of the photodiode.

TRANSIMPEDANCE AMPLIFIER



Place caps as close to the V+ terminal of the amp as possible

AMPLIFIER SWITCHING



K1 and K2 allow photocurrent to reach the sergeant channel of the ACF integrator, which can give time-averaged photocurrent measurements for both the sergeant and the soldier photodiodes with nearly identical characteristics between the two.

K3 and K4 allow photocurrent to reach the IVC integrator, which serves to validate the value read from the sergeant channel of the ACF.

The relays are paired and wired in series (such that the switch terminals are all Pin 3) for extremely low EMF

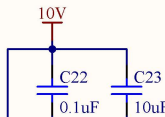
Pickering 118-A-5/2D reed relays are used for much lower leakage current than solid-state switches while taking very little board space for relays.

Relay coil resistance (i.e., between A5VSW and MOSFET drains) is 2.2k per relay (1.1k per pair).

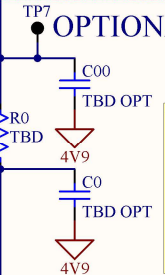
BIAS VOLTAGE CONFIGURATION

DAC U9 sets SPI-controlled bias voltage for photodiode reverse-bias operation. Nominal 2.5V output. Current limit 10mA.
AD5601 is a drop-in replacement for DAC5311 with slightly worse noise and linearity but identical pinout and interface. DAC5311 is preferred if available.

Per the AD5601 datasheet, C23 should be a tantalum capacitor. It is currently specified as an MLCC.

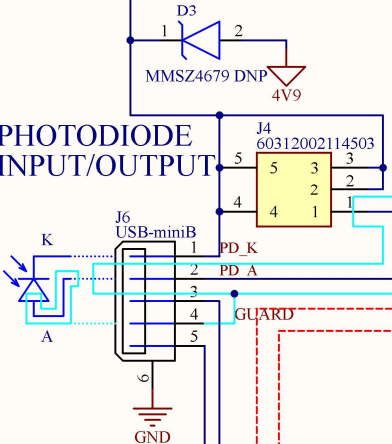


OPTIONAL RC FILTER



If used, place RC filter as close as possible to the photodiode anode.
Otherwise, DNP the capacitors and set R0 = 0 Ohms.

PHOTODIODE INPUT/OUTPUT

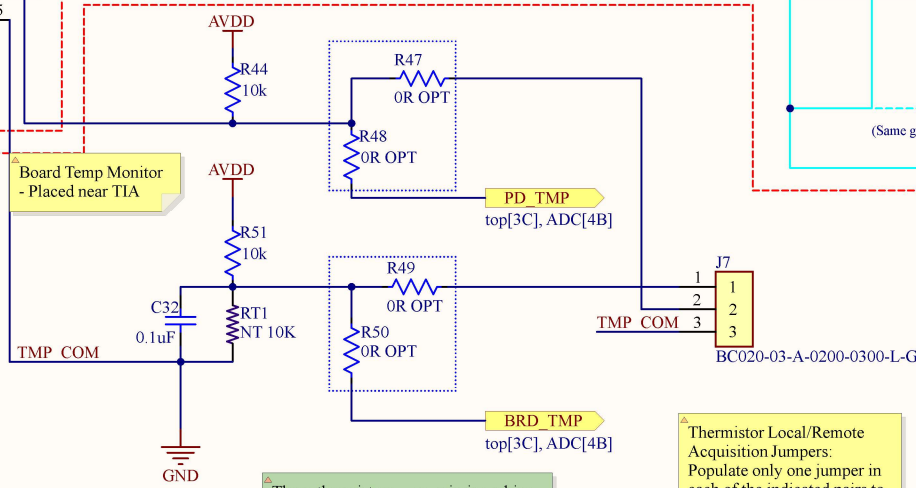


D3 Photodiode protection Zener diode. Should only be placed if software cannot be trusted to operate the DAC safely. In that case, it should be populated with a Zener that will break down before the photodiode does, and paired with a current-limiting resistance in the RC filter. Most Zeners will have microamp scale leakage current. This acts as an additional source of shot noise that adds together with that from the photodiode dark current and limits measurement precision.

Here, we expect the anode to be at 4.9V and the cathode to be somewhere between 4.9V and the maximum reverse-bias voltage.

U12 is a 6-channel digital isolator. This is necessary to propagate the DAC SPI and bias switch control to the virtual-ground components.
The A5VSW should arguably be DVDD, since 3.3V is a bit under 0.7*5V and is thus slightly out of spec.
A 6-channel isolator is necessary for the integrator controls, so a second one is used here with 2 unused channels instead of using a 4-channel variant. The 4-channel isolators are typically no smaller than 6-channel isolators.

TEMPERATURE MONITORING



These thermistors were missing a bias source so wouldn't output anything. We use the nice, clean AVDD for this now

Thermistor Local/Remote Acquisition Jumpers: Populate only one jumper in each of the indicated pairs to use on-board or off-board ADC (not both).

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Title ALTAIR Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)		
Size	Number	Revision
Tabloid		V26C (V26.4a)
Date:	6-03-2026	Sheet 4 of 6
File:	C:\Users\U.TIA.SchDoc	Drawn By: Nicholas Sandor

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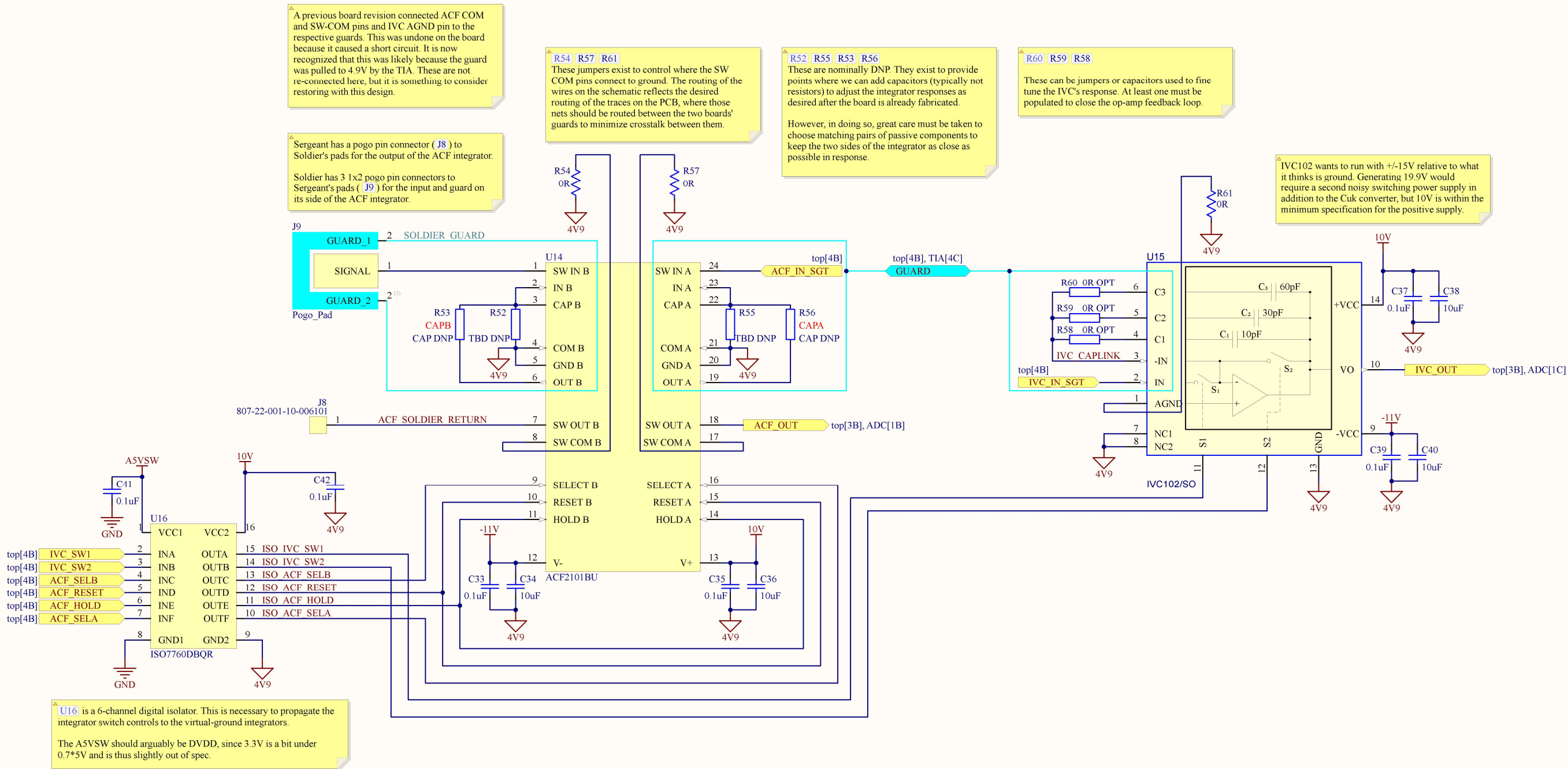
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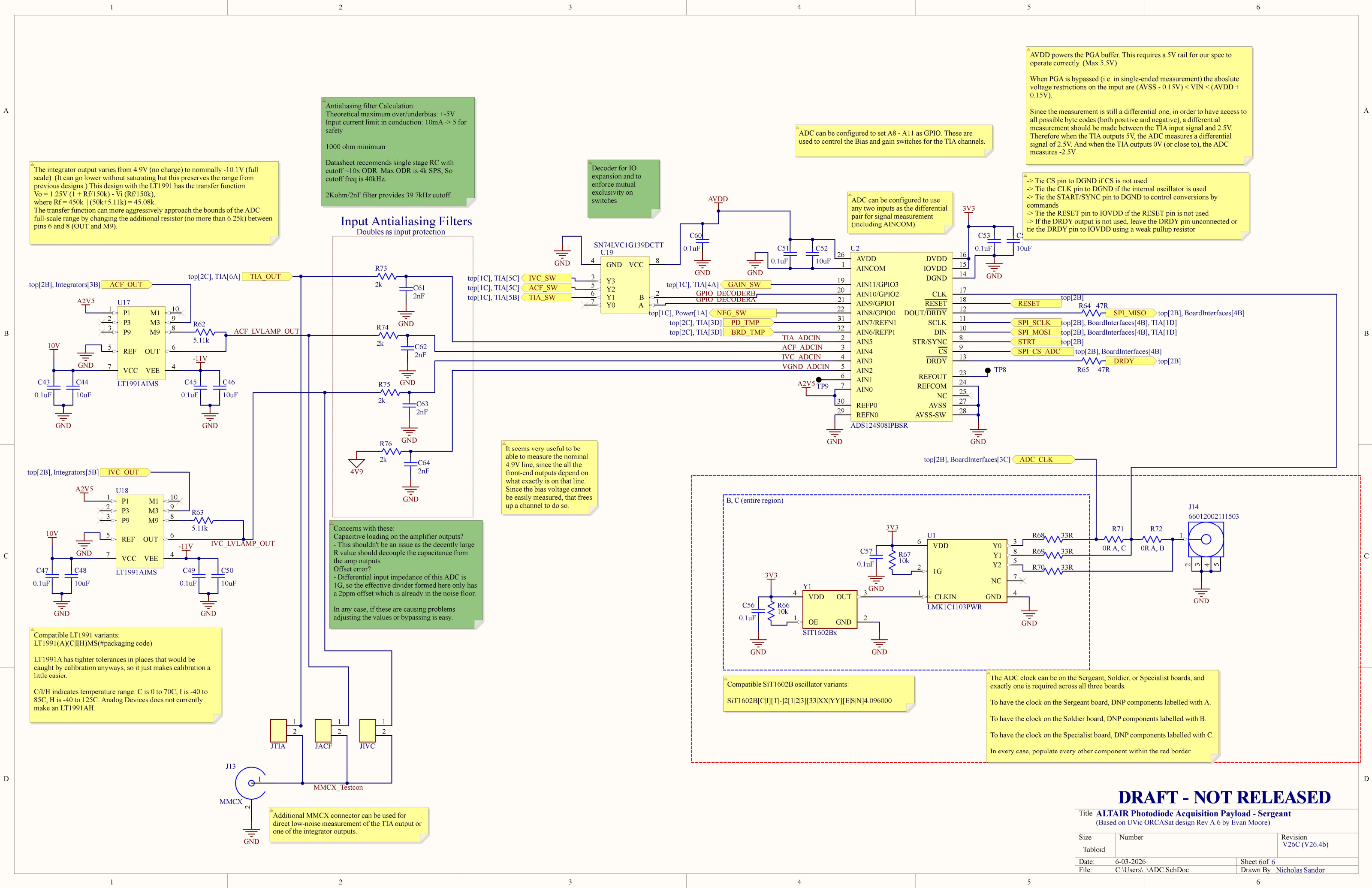
OPERATING THE ACF2101		
FUNCTION	HOLD SWITCH	RESET SWITCH
INTEGRATE	CLOSED	OPEN
HOLD	OPEN	OPEN
RESET	OPEN	CLOSED

OPERATING THE IVC102		
FUNCTION	SWITCH 1	SWITCH 2
INTEGRATE	CLOSED	OPEN
HOLD	OPEN	OPEN
RESET	OPEN	CLOSED

Switches are TTL/CMOS compatible.
Switch Open: Logic HIGH (3.3V or 5.0V)
Switch Closed: Logic LOW (0V)

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Title ALTair Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)		
Size Tabloid	Number	Revision V26C (V26.4b)
Date:	6-03-2026	Sheet 5 of 6
File:	C:\Users\...\Integrators.SchDoc	Drawn By: Nicholas Sandor



The integrator output varies from 4.9V (no charge) to nominally -10.1V (full scale). (It can go lower without saturating but this preserves the range from previous designs.) This design with the LT1991 has the transfer function $V_o = 1.25V (1 + R_f/150k) - V_i (R_f/150k)$, where $R_f = 450k \parallel (50k + 5.11k) = 45.08k$. The transfer function can more aggressively approach the bounds of the ADC full-scale range by changing the additional resistor (no more than 6.25k) between pins 6 and 8 (OUT and M9).

Antialiasing filter Calculation:
Theoretical maximum over/underbias: +5V
Input current limit in conduction: 10mA -> 5 for safety

1000 ohm minimum

Datasheet recommends single stage RC with cutoff ~10x ODR. Max ODR is 4k SPS, So cutoff freq is 40kHz.

2Kohm/2nF filter provides 39.7kHz cutoff.

Input Antialiasing Filters

Doubles as input protection

Decoder for IO expansion and to enforce mutual exclusivity on switches

ADC can be configured to set A8 - A11 as GPIO. These are used to control the Bias and gain switches for the TIA channels.

AVDD powers the PGA buffer. This requires a 5V rail for our spec to operate correctly. (Max 5.5V)

When PGA is bypassed (i.e. in single-ended measurement) the absolute voltage restrictions on the input are $(AVSS - 0.15V) < V_{IN} < (AVDD + 0.15V)$.

Since the measurement is still a differential one, in order to have access to all possible byte codes (both positive and negative), a differential measurement should be made between the TIA input signal and 2.5V. Therefore when the TIA outputs 5V, the ADC measures a differential signal of 2.5V. And when the TIA outputs 0V (or close to), the ADC measures -2.5V.

-> Tie CS pin to DGND if CS is not used
-> Tie the CLK pin to DGND if the internal oscillator is used
-> Tie the START/SYNC pin to DGND to control conversions by commands
-> Tie the RESET pin to IOVDD if the RESET pin is not used
-> If the DRDY output is not used, leave the DRDY pin unconnected or tie the DRDY pin to IOVDD using a weak pullup resistor

ADC can be configured to use any two inputs as the differential pair for signal measurement (including AINCOM).

It seems very useful to be able to measure the nominal 4.9V line, since the all the front-end outputs depend on what exactly is on that line. Since the bias voltage cannot be easily measured, that frees up a channel to do so.

Concerns with these:
Capacitive loading on the amplifier outputs?
- This shouldn't be an issue as the decently large R value should decouple the capacitance from the amp outputs
Offset error?
- Differential input impedance of this ADC is 1G, so the effective divider formed here only has a 2ppm offset which is already in the noise floor.

In any case, if these are causing problems adjusting the values or bypassing is easy.

Compatible LT1991 variants:
LT1991(A)(C)(I)(H)MS(#packaging code)

LT1991A has tighter tolerances in places that would be caught by calibration anyways, so it just makes calibration a little easier.

C/I/H indicates temperature range. C is 0 to 70C, I is -40 to 85C, H is -40 to 125C. Analog Devices does not currently make an LT1991AH.

Compatible SiT1602B oscillator variants:
SiT1602B[C](I)[T]-[2][1][2][3][33][XX][YY][E][S][N]4.096000

The ADC clock can be on the Sergeant, Soldier, or Specialist boards, and exactly one is required across all three boards.

To have the clock on the Sergeant board, DNP components labelled with A.

To have the clock on the Soldier board, DNP components labelled with B.

To have the clock on the Specialist board, DNP components labelled with C.

In every case, populate every other component within the red border.

Additional MMCX connector can be used for direct low-noise measurement of the TIA output or one of the integrator outputs.

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Title ALTAIR Photodiode Acquisition Payload - Sergeant (Based on UVic ORCASat design Rev A.6 by Evan Moore)		
Size Tabloid	Number	Revision V26C (V26.4b)
Date:	6-03-2026	Sheet 6 of 6
File:	C:\Users\...\ADC.SchDoc	Drawn By: Nicholas Sandor